Arash Ahmadi

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Education

- **Doctor of Philosophy (Ph.D.), Electrical and Computer Engineering** — Aug 2024 – Current
  University of Oklahoma, Norman, OK, USA
  Research focus: Fine-tuning small language models for efficient deployment on resource-constrained edge devices, with applications in agentic systems and diverse domains.
- **Master of Science (M.S.), Electrical and Computer Engineering** — Aug 2024 – Spring 2026
  University of Oklahoma, Norman, OK, USA
  GPA: 3.88/4.0
  Thesis: *Search-Based Reward Function Optimization for LLM Reasoning via Reinforcement Learning*
- **Bachelor of Computer Engineering, Software Engineering** — Sep 2018 – Feb 2023
  University of Kurdistan, Sanandaj, Kurdistan, Iran
  GPA: 3.88/4.0
  Thesis: *Efficient brute-force state space search for Yin-Yang puzzle*

Publications

Journal Articles

[J1] **A. Ahmadi**, S. S. Sharif, and Y. M. Banad, "Improving Aviation Safety Analysis: Automated HFACS Classification Using Reinforcement Learning with Group Relative Policy Optimization," *Expert Systems with Applications*, Article 132963, 2026. doi:10.1016/j.eswa.2026.132963.

[J2] **A. Ahmadi**, S. S. Sharif, and Y. M. Banad, "A comparative study of sampling methods with cross-validation in the fedhome framework," *IEEE Transactions on Parallel and Distributed Systems*, vol. 36, no. 3, pp. 570-579, 2025. doi:10.1109/TPDS.2025.3526238.

[J3] **A. Ahmadi** and A. Khorramian, "Efficient Brute-force state space search for Yin-Yang puzzle," *The Journal of Supercomputing*, vol. 80, no. 3, pp. 3066-3088, 2024. doi:10.1007/s11227-023-05565-w.

Conference Proceedings

[C1] **A. Ahmadi**, S. S. Sharif, and Y. M. Banad, "LLM-Powered HFACS Analysis of ASRS Narratives via MCP Bridge for Enhanced Aviation Safety Debriefs," in *AIAA SCITECH 2026 Forum*, p. 1195, Jan 2026. doi:10.2514/6.2026-1195.

[C2] **A. Ahmadi**, S. Sharif, and Y. M. Banad, "Towards Transparent Artificial Intelligence: Exploring Modern Approaches and Future Directions," in *2024 Conference on AI, Science, Engineering, and Technology (AIxSET)*, pp. 248-251, IEEE, 2024. doi:10.1109/AIxSET62544.2024.00047.

Preprints & Under Review

[P1] **A. Ahmadi**, S. S. Sharif, and Y. M. Banad, "Enhanced LLM Reasoning by Optimizing Reward Functions with Search-Driven Reinforcement Learning," *arXiv preprint arXiv:2605.02073*, 2026. doi:10.48550/arXiv.2605.02073.

[P2] **A. Ahmadi**, S. Sharif, and Y. M. Banad, "MCP Bridge: A Lightweight, LLM-Agnostic RESTful Proxy for Model Context Protocol Servers," *arXiv preprint arXiv:2504.08999*, 2025. doi:10.48550/arXiv.2504.08999.

Intellectual Property

- **A. Ahmadi**, Y. M. Banad, and S. Sharif. "Artificial Intelligence-Powered Personal Computer Management System and Methods," U.S. Nonprovisional Patent Application No. 19/378,506, filed November 4, 2025.

Research Experience

- **Graduate Research Assistant**, Inquire Lab — Aug 2024 – Present
  University of Oklahoma, Norman, OK, USA (Supervisors: Dr. Yaser Mike Banad, Dr. Sarah Sharif)
  - My research focuses on optimizing and fine-tuning language models for deployment on resource-constrained edge devices. This work explores their applications in agentic systems and diverse domains such as aviation safety and health monitoring.
- **Research Assistant** — Feb 2023 – Aug 2024
  University of Kurdistan (Supervisor: Dr. Amanj Khorramian)
  - Conducted research on theoretical computer science projects in the fields of Graphs, Puzzles, and Parallel Algorithms.
  - Proposed, implemented, and parallelized a novel algorithm for the Yin-Yang puzzle; this work included analyzing the results and authoring the majority of the publication.

Awards and Honors

- William H. Barkow Graduate Fellowship, University of Oklahoma 2025-2026
- William H. Barkow Scholarship, University of Oklahoma 2025-2026
- Journal Paper Award, Electrical and Computer Engineering, University of Oklahoma 2025
  For the publication: "A comparative study of sampling methods with cross-validation in the fedhome framework"

Teaching and Mentoring

- **Instructor and Lead Developer** 2025, 2026
  Cybersecurity Essentials Workshop, University of Oklahoma
  - Led a 10-person team in developing and delivering a five-day intensive workshop for IT professionals, sponsored by the Oklahoma Office of Homeland Security, Oklahoma Homeland Security, and FEMA.
  - Designed the 2025 workshop agenda and description, shaping the instructional flow, topics, and participant-facing materials.
  - Authored and presented modules on emerging cyber threats, ransomware prevention, phishing analysis, secure network design, and firewall best practices.
  - Designed and developed the official workshop website and a custom content management platform.
- **Teaching Assistant** Fall 2019 – Spring 2020
  University of Kurdistan
  - Courses: Advanced Programming with Java (Spring 2020), Fundamentals of Programming in C (Fall 2019).
  - Conducted weekly review sessions and provided direct feedback on projects and assignments.

Professional and Community Service

- **Ad hoc Reviewer**, *IEEE Transactions on Artificial Intelligence* 2026
- **Volunteer**, The Big Event, University of Oklahoma 2026
- **Volunteer**, The Little Event, University of Oklahoma Oct 2026

Selected Technical Projects

- **Inquire Lab Research Website (inquirelab.ai):** Designed and developed the official website for the Inquire Lab with Next.JS. The site serves as a public-facing platform to showcase research projects, publications, and team members.
- **Traffic Prediction in Edge Environments (Python, 2021):** Implemented a federated learning approach with a GRU network in TensorFlow to predict traffic. This method enhanced privacy and reduced communication costs.
- **Task Scheduling Optimization (Python, 2020):** Developed an application with a UI to implement and compare heuristic search algorithms (e.g., Greedy, Hill-climbing, Genetic Algorithm) for task scheduling.

Leadership and Activities

- **Founder**, LLM Engineering Club, University of Oklahoma Mar 2026 – Present
  - Founded and organized a registered student organization for hands-on LLM engineering, responsible AI practice, workshops, study groups, and collaborative AI projects.
  - Organized club programming around lab-approved access to a computing cluster for local model training, fine-tuning, and evaluation.
  - Led AI news discussions and designed a workshop series where students train a small language model to play a Codenames-style word-reasoning game.
- **Engineering Officer**, Game Developer's Association, University of Oklahoma Aug 2024 – Dec 2024
  - Delivered technical presentations on engineering principles in video game development.
  - Developed an interactive Ouija board powered by the Gemini 2.0 Flash large language model for a university-wide Halloween escape room event.

References

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